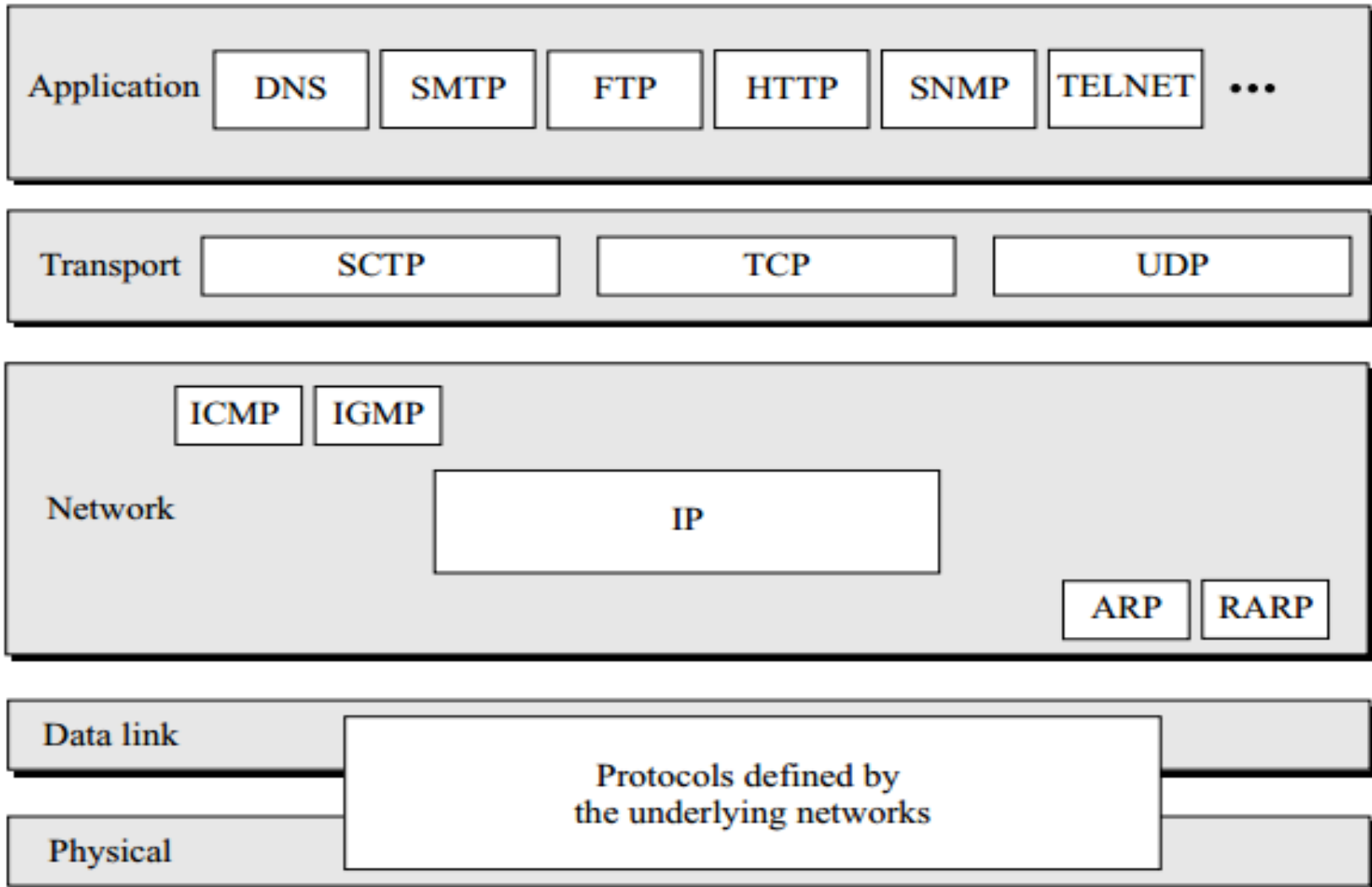


TCP/IP Protocol Suite

TCP/IP Protocol Suite

- The networking model used in the Internet today is the **Transmission Control Protocol/Internetworking Protocol** (TCP/IP) or TCP/IP Protocol Suite.
- TCP/IP is a **hierarchical** protocol made up of interactive modules:
 - application, transport, network, data link, and physical

TCP/IP Protocol Suite



Application Layer

- The application layer is responsible for providing services to the user.
- It provides user interfaces and support for services such as file transfer, electronic mail, and remote logging.

Application Layer Protocols

- Domain Name System (DNS)
 - It finds the logical (network-layer) address when given the specific (application-layer) address.
- Simple Mail Transfer Protocol (SMTP)
 - used for electronic mail
- File Transfer Protocol (FTP)
 - used to transfer large files from one computer to another.
- Hypertext Transfer Protocol (HTTP)
 - used to access the World Wide Web (WWW).

Application Layer Protocols

- Simple Network Management Protocol (SNMP)
 - official management protocol in the Internet.
- Terminal Network (TELNET)
 - remote log-in application program.
 - A user can use TELNET to connect to a remote host and use the available services.

Transport Layer

- The transport layer is responsible for the delivery of a message from one process to another.
- Traditionally the transport layer was represented in TCP/IP by two protocols: **TCP** and **UDP**.
- **Stream Control Transmission Protocol (SCTP)** provides support for new applications such as IP telephony.

Transport Layer Protocols

- User Datagram Protocol (UDP)
 - adds only port addresses, checksum error control, and length information to the data from the upper layer.
- Transmission Control Protocol (TCP)
 - Provides full transport layer services to applications.
 - TCP is a **reliable** (connection-oriented) **stream** transport protocol.
 - TCP divides a stream of data into smaller units called segments.
 - Each segment includes a sequence number for reordering after receipt, together with an acknowledgment number for the segments received.

Network Layer

- The network layer is responsible for the delivery of individual packets from the source host to the destination host.
 - Provides logical addressing and routing

Network Layer Protocols

- Internet Protocol (IP)
 - unreliable and connectionless protocol
 - provides no error checking or tracking
 - transports data in packets called **datagrams**, each of which is transported separately.
 - IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination.

Network Layer Protocols

- Address Resolution Protocol (ARP)
 - used to associate an IP address with the physical address.
- Reverse Address Resolution Protocol (RARP)
 - allows a host to discover its Internet address when it knows only its physical address.

Network Layer Protocols

- Internet Control Message Protocol (ICMP)
 - sends query and error reporting messages
 - is a mechanism used by hosts and other intermediate devices to send notification of datagram problems
- Internet Group Message Protocol (IGMP)
 - used to facilitate the simultaneous transmission of a message to a group of recipients.

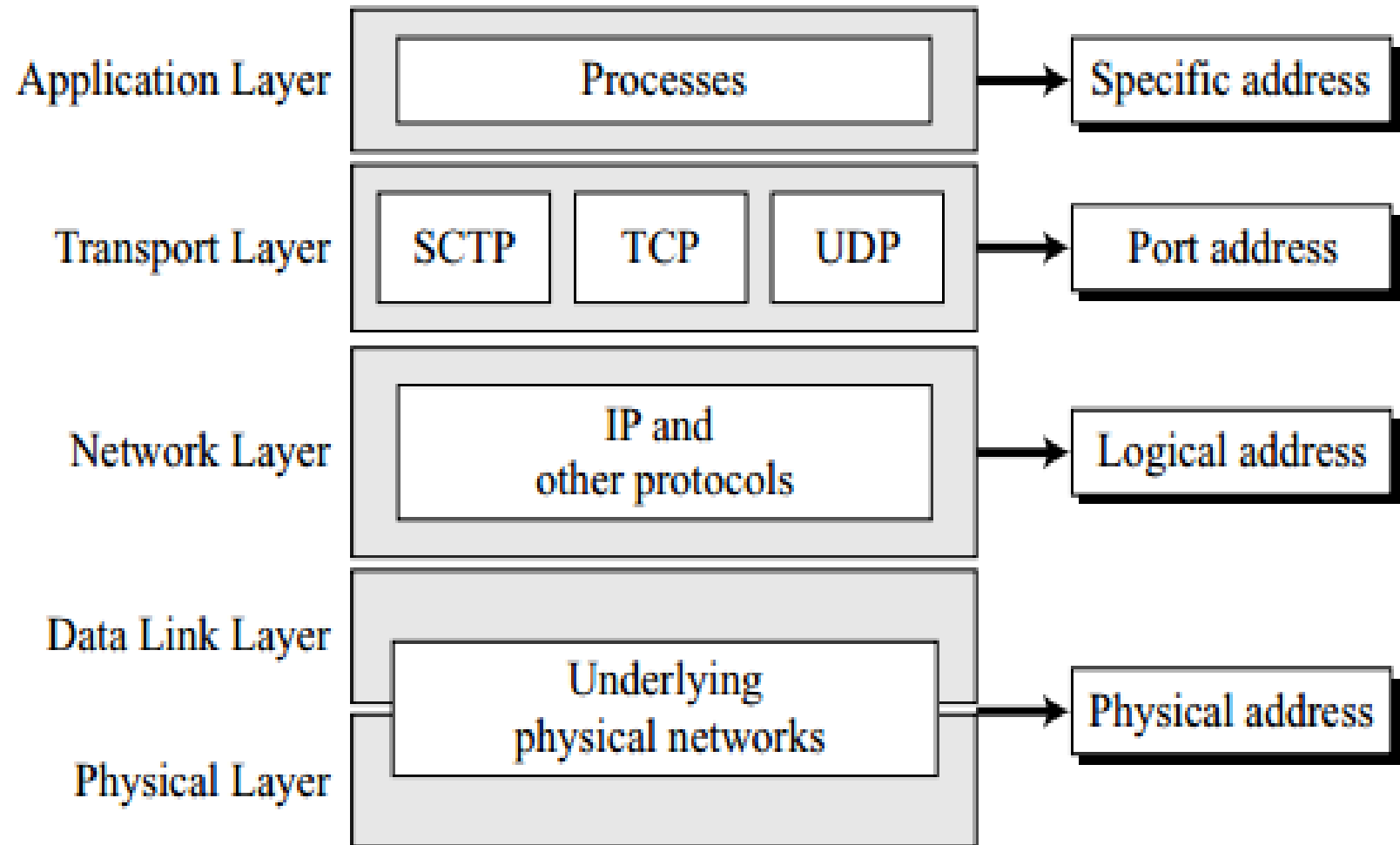
Data Link Layer

- The data link layer is responsible for moving **frames** from one hop (node) to the next.
- It transforms the physical layer, a raw transmission facility, to a reliable link
- It makes the physical layer appear error-free to the upper layer
- Some responsibilities are **framing, physical addressing, flow control, error control, and access control.**

Physical Layer

- The physical layer is responsible for movements of individual bits from one hop (node) to the next.
- Concerned with:
 - physical characteristics of interfaces and transmission media
 - representation of bits, data rate, synchronization of bits
 - physical topology

Addressing in TCP/IP protocols



Addressing in TCP/IP protocols

- Specific addresses
 - addresses belonging to specific application layer protocols
- Port address:
 - the label assigned to a process
- Logical addresses
 - necessary for universal communication services that are independent of underlying physical networks
 - A logical address (IP address) in the Internet is currently a 32-bit address
- Physical Address
 - also known as the link address
 - It is the address of a node as defined by its physical network

Security Protocols

- Security at Application Layer:
- Two protocols providing security services for e-mails:
 - Pretty Good Privacy (PGP)
 - Secure/Multipurpose Internet Mail Extension (S/MIME)

Security Protocols

- Security at Transport Layer:
 - Secure Socket Layer (SSL)
 - Transport Layer Security (TLS)
- Security at Network Layer:
 - IPSec